

Golf Cart Diagnostic Technician Training Program

Complete Ready-to-Deliver Package · Version 1.1 · July 2026

Format: 10 weeks · 20 sessions · 40 contact hours (2 hrs × 2 sessions/week)

Delivery mix: ≥60% hands-on lab · Classroom theory + shop practice

Audience: Golf courses, resorts, fleet operators, independent technicians

Package Contents

Folder / File	Contents
00_program_guide.md	Learning objectives map, rubrics, class size/ratio, scaling, customization
00_instructor_master_checklist.md	Cohort-wide staging, weekly rhythm, hands-on ratio audit
weeks/	Weeks 1–10: timed lesson plans (both sessions), quizzes + keys, instructor prep
labs/	Printable lab worksheets, checklists, data logs, diagnostic report forms
handouts/	Student references (symbols, flowcharts, PM, Curtis, Ohm's Law, tools, resources)
final_assessment/	Practical rubric, 50-question exam + key, certificate, grade tracker
ITERATION_LOG.md	Self-improvement review cycles against delivery goals

Quick Start for Instructors

1. Read **Program Guide** end-to-end once before Week 1.
2. Print/copy **labs/** and **handouts/** for the student binder.
3. Each week: follow that week's file — Session 1 (theory), Session 2 (lab).
4. Use **Instructor Preparation** section at the top of each week file the day before.
5. Week 10: run final practical + written exam from [final_assessment/](#).

GitHub Pages (Command Center)

After merge to main (Pages deploy):

- **Training Hub:** <https://ngc4160.github.io/NGC-Brain/command-center/training/>
- **Command Center home:** <https://ngc4160.github.io/NGC-Brain/command-center/>
- Login with the shared NGC access code (Ryan / Christine).

Browse weeks, labs, handouts, and finals in-browser via the hub or Full Index → **Technician Training**.

Master Schedule at a Glance

Week	Theme	Session 1	Session 2
1	Safety, Tools & Electrical Fundamentals	Theory + pre-assessment	Multimeter / LOTO lab
2	Lead-Acid Battery Systems	Battery theory & hazards	Full battery diagnostic lab
3	Chargers & Lithium	Charger stages + BMS intro	Charger & lithium diagnostics
4	Electric Motors	Motor types & wear	Motor inspection & electrical tests
5	Controllers, Throttle & Solenoids	PWM, Curtis, HPD	Fault codes & component tests
6	Wiring Diagrams & Schematics	Symbols & brand harnesses	Tracing & planted faults
7	Diagnostic Methodology	7-step process	Live-cart applied diagnostics
8	Mechanical Systems	Brakes, steering, drivetrain	Inspection & road test
9	Gas-Powered Carts	Engine & fuel/ignition	Compression, spark, fuel checks
10	PM, Case Studies & Certification	PM programs & careers	Presentations + finals + certificates

Safety Emphasis (Non-Negotiable)

Post and enforce Appendix-style shop rules from handouts. Every session opens with a 2–5 minute safety reminder. Treat every pack as energized until verified zero energy with a meter.

Brand & Tech Coverage

Club Car · EZ-GO (Textron) · Yamaha · Curtis controllers (1204/1205/1268 family and similar) · Alltrax (overview) · Lead-acid & lithium (LiFePO₄) · Gas single-cylinder 4-stroke

Diagram Placeholders

Where diagrams are needed, files include clearly labeled placeholders in the form:

[DIAGRAM PLACEHOLDER: ...] — description of what to draw/photo later

Preserve “diagnose before replace.” Update annually for new models and lithium tooling.